ARTIFICIAL INTELLIGENCE — DETAILED MASTER INDEX

Detailed Master Index • B.Tech CSE 4th Year • Placement / Interview / OA Preparation

0. COMMON CSE + INTERVIEW FOUNDATION

0.1 Programming [B][INT][OA] — Variables, data types, operators, input/output, conditionals, loops, functions, scope, recursion, exceptions and debugging.

0.2 Python [B][INT][OA] — Lists, tuples, sets, dictionaries, strings, slicing, comprehensions, lambda, map/filter/reduce, iterators, generators and modules.

0.3 OOP [B][INT] — Class/object, constructor, encapsulation, inheritance, polymorphism, abstraction, composition, overriding, overloading concepts and SOLID basics.

0.4 DSA [B→A][INT][OA] — Arrays, strings, hashing, linked lists, stacks, queues, trees, BST, heap, trie, graphs, recursion, backtracking and dynamic programming.

0.5 Algorithms [B→A][INT][OA] — Sorting, binary search, two pointers, sliding window, prefix sums, greedy, divide-and-conquer, BFS/DFS and DP.

0.6 Complexity [B][INT][OA] — Big-O, Big-Theta, Big-Omega, time/space complexity, recursion complexity and amortized analysis.

0.7 SQL [B→A][INT][OA] — SELECT, WHERE, GROUP BY, HAVING, joins, subqueries, CTEs, CASE, windows, ranking, dates, NULLs and query optimization.

0.8 Git/Linux [B][P] — Git workflow, branches, merge/rebase, GitHub, CLI, permissions, processes, pipes, grep/sed/awk and environment variables.

0.9 Interview method [INT] — Clarify → approach → brute force → optimize → code → test edge cases → complexity → alternatives → trade-offs.

1. AI FUNDAMENTALS

1.1 AI overview [B][INT] — Definitions, goals, history, narrow/general/super AI, applications and limitations.

1.2 Intelligent agents [B][INT] — Environment, sensors, actuators, rationality, PEAS and agent architectures.

1.3 Problem solving [B→I] — State space, initial/goal states, actions, path cost and problem formulation.

1.4 Uninformed search [I][INT] — BFS, DFS, depth-limited, iterative deepening and uniform-cost search.

1.5 Informed search [I][INT] — Heuristics, greedy best-first, A*, admissibility and consistency.

1.6 Adversarial search [I] — Minimax, game trees, alpha-beta pruning and evaluation functions.

1.7 Knowledge representation [I] — Propositional logic, first-order logic, rules, semantic networks, frames and knowledge graphs.

1.8 Reasoning [I→A] — Inference, uncertainty, Bayesian reasoning, decision theory and probabilistic models.

1.9 AI ethics [I][INT] — Bias, fairness, privacy, explainability, safety, accountability and responsible AI.

2. MATHEMATICS FOR AI [M]

2.1 Linear algebra — Scalars, vectors, matrices, matrix operations, dot products, norms, projections, rank, eigenvalues/eigenvectors and SVD concepts.

2.2 Probability — Sample space, conditional probability, independence, Bayes theorem, random variables, expectation and variance.

2.3 Distributions — Bernoulli, binomial, categorical, normal, Poisson and exponential distributions.

2.4 Statistics — Mean, median, variance, covariance, correlation, sampling, confidence intervals and hypothesis basics.

2.5 Calculus — Derivatives, partial derivatives, gradients, chain rule, Jacobian/Hessian concepts and optimization.

2.6 Optimization — Objective functions, local/global minima, convexity intuition, gradient descent and learning-rate behavior.

3. MACHINE LEARNING CORE [B→A][INT][OA]

3.1 ML concepts — Supervised, unsupervised, semi-supervised and reinforcement learning; features, labels and objectives.

3.2 Data preparation — Cleaning, missing values, outliers, encoding, scaling, feature engineering, train/validation/test split and leakage.

3.3 Regression — Linear, polynomial, Ridge, Lasso and Elastic Net.

3.4 Classification — Logistic regression, k-NN, Naive Bayes, decision trees and SVM.

3.5 Ensembles — Bagging, random forest, boosting, AdaBoost, gradient boosting, XGBoost and LightGBM concepts.

3.6 Unsupervised learning — K-means, hierarchical clustering, DBSCAN, PCA and anomaly detection.

3.7 Evaluation — MAE, MSE, RMSE, R^2, confusion matrix, precision, recall, F1, ROC-AUC, PR-AUC and calibration.

3.8 Generalization — Bias/variance, underfitting, overfitting, regularization and cross-validation.

4. DEEP LEARNING [I→A][INT]

4.1 Neural networks — Perceptron, layers, weights, biases, forward propagation and computation graphs.

4.2 Activations — Sigmoid, tanh, ReLU, Leaky ReLU, GELU and softmax.

4.3 Loss functions — MSE, binary cross-entropy, categorical cross-entropy and contrastive-loss concepts.

4.4 Backpropagation — Chain rule, gradients, gradient descent and computational graphs.

4.5 Training — Batch/mini-batch/SGD, initialization, learning rate, epochs and batching.

4.6 Optimizers — SGD, momentum, RMSProp, Adam and AdamW concepts.

4.7 Regularization — Dropout, weight decay, early stopping and batch normalization.

4.8 CNN — Kernels, convolution, stride, padding, feature maps, pooling, receptive fields and image applications.

4.9 RNN family — Sequence modeling, BPTT, vanishing gradients, LSTM and GRU.

4.10 Transfer learning — Pretraining, feature extraction, fine-tuning and parameter-efficient approaches.

5. GENERATIVE AI + LLMs [I→A][INT]

5.1 Generative modeling — Autoregressive models, autoencoders, VAEs, GANs and diffusion overview.

5.2 Transformers — Tokens, embeddings, Q/K/V, attention, self-attention, multi-head attention and positional encoding.

5.3 Transformer architecture — Encoder, decoder, masked attention, residual connections, normalization and feed-forward layers.

5.4 LLM fundamentals — Tokenization, next-token prediction, pretraining, context window and inference.

5.5 Prompt engineering — Zero-shot, few-shot, role/context, structured output, constraints and prompt evaluation.

5.6 Embeddings — Semantic vectors, similarity, cosine distance and embedding models.

5.7 RAG — Ingestion, chunking, embedding, indexing, retrieval, reranking, context assembly and grounded generation.

5.8 Fine-tuning — Instruction tuning, LoRA/PEFT concepts, datasets, evaluation and when to fine-tune.

5.9 Agents — Tools, planning, memory, workflows, function calling, guardrails and evaluation.

5.10 LLM evaluation/safety — Hallucination, groundedness, relevance, prompt injection, data leakage, toxicity and monitoring.

6. AI ENGINEERING + MLOPS [A][P][INT]

6.1 Python AI stack — NumPy, pandas, scikit-learn, PyTorch/TensorFlow concepts and experiment notebooks.

6.2 APIs — REST, JSON, authentication, validation, error handling and model inference endpoints.

6.3 Databases — SQL, NoSQL, object storage and vector database concepts.

6.4 Deployment — Docker, environments, model serving, batch vs real-time inference and cloud basics.

6.5 MLOps — Experiment tracking, model registry, pipelines, versioning, reproducibility and CI/CD.

6.6 Production — Latency, throughput, batching, caching, scaling, cost, reliability, observability and security.

7. AI PROJECTS + INTERVIEW/OA [P][INT][OA]

7.1 Classical AI project — Search/planning/recommendation problem with algorithm comparison and evaluation.

7.2 ML project — Dataset → preprocessing → baseline → model → tuning → evaluation → API.

7.3 Deep learning project — Training pipeline → experiments → error analysis → deployment.

7.4 GenAI project — LLM/RAG application with retrieval, prompting, evaluation and UI/API.

7.5 AI system case study — Requirements → data → model → inference → monitoring → safety → cost.

7.6 Interview bank — AI definitions, algorithm comparisons, ML theory, DL theory, LLM/RAG questions and project defense.

7.7 OA bank — DSA, Python, SQL, probability/statistics, ML MCQs and timed coding.

Final preparation sequence: Fundamentals → Core theory → Implementation → Problem solving → Projects → Deployment → Technical interview → OA → Mock interview → Revision.